

PCB Materials & Substrates

FR-4 • Rogers • Polyimide • Megtron • Ceramic — choosing the right base

The dielectric decides everything: cost, frequency, temperature, reliability

Audience: Intermediate Engineering Students

Substrate types • Tg • Dk/Df • Cost • Selection

Why Material Choice Matters

Substrate dielectric drives PCB performance + cost

Cost

FR-4 ≈ ₹100/board
Rogers ≈ ₹1000/board
Ceramic ≈ ₹10k/board

Frequency

FR-4 capped at ~3 GHz
Rogers good to 40 GHz
Ceramic to 100 GHz+

Temperature

FR-4 T_g 130-180 °C
Polyimide to 260 °C
Ceramic 1000+ °C

Thermal cond.

FR-4 0.3 W/mK
Metal-core 1-3 W/mK
Ceramic 25-180 W/mK

Reliability

FR-4 = mass market
Polyimide = mil/aero
Ceramic = space, RF

Manufacturability

FR-4 every fab
Rogers select fabs
Ceramic specialists

FR-4 Grades

Tg distinguishes the variants you'll buy

FR-4 grade	Tg (°C)	Td (°C)	Use case	Cost mult.
Standard FR-4	130-140	300	Consumer, low-cost	1.0× (baseline)
FR-4 Medium-Tg	150-160	320	Industrial, automotive	1.2×
FR-4 High-Tg	170-180	340	Lead-free assembly, multilayer	1.5×
FR-4 Ultra-High-Tg	180+	360	Heavy lead-free, mil-spec	2.0×
Halogen-Free FR-4	130-180	300+	RoHS-strict, EU enclosed	1.3×
FR-4 with low Df	150-170	340	Sub-6 GHz RF	1.5-2×

Tg = Glass Transition Temperature

Above Tg, the resin softens — copper/glass adhesion drops. Pick Tg ≥ peak reflow temperature + margin. For SAC305 reflow (245 °C peak), use ≥ 170 °C Tg.

Resin Systems

Beyond FR-4 — when standard epoxy isn't enough

Epoxy (FR-4)

Workhorse — cheap, easy.
Capped at ~3 GHz.

Polyimide

High-temp (260 °C cont.)
Flex PCBs, aerospace.

Cyanate Ester (BT)

Low CTE, BGA substrates.
Used in IC packages.

PPE / PPO blends

Low loss + low Dk.
DDR4, 10G Ethernet.

Hydrocarbon (Rogers RO4000)

Sub-6 GHz RF.
Mid-cost between FR-4 + PTFE.

PTFE / Teflon

Very low loss + low Dk.
mmWave, radar, satellite.

Rogers Corporation Materials

The industry standard for RF / mmWave PCBs

Series	Dk	Df	Max freq	Cost vs FR-4
RO4350B	3.48	0.0037	6 GHz	5-8×
RO4003C	3.38	0.0021	10 GHz	8-10×
RO3003	3.0	0.0013	24-77 GHz radar	10-15×
RO3035	3.5	0.0015	30 GHz	10-15×
RT/duroid 5880	2.2	0.0009	100 GHz	15-20×
RT/duroid 6002	2.94	0.0012	60 GHz	12-18×

Pick Rogers when

Operating frequency > 3 GHz, OR insertion loss matters, OR temperature-stable Dk is required. Below 3 GHz, FR-4 with careful design works fine and saves money.

Polyimide (Kapton)

Flex PCBs, high-temp applications

- ▶ Polyimide film (DuPont Kapton) — substrate for flex circuits
- ▶ Continuous use up to 260 °C — survives reflow + automotive
- ▶ $T_g > 360$ °C — essentially no glass-transition concern
- ▶ $Dk \approx 3.5$, $Df \approx 0.008$ — decent RF (better than FR-4)
- ▶ Available with Cu foil + adhesive (3-layer) or adhesive-less (2-layer)
- ▶ Cost: 3-5× FR-4 by area, but enables much smaller / flexible designs

Where polyimide wins

- Phone hinges & cameras (flex tail)
- Aerospace cable harnesses
- Wearables / smart clothes
- Medical implants
- Automotive (engine bay, sensor lines)

Combined with FR-4

Rigid-flex constructions bond polyimide flex tails to FR-4 rigid sections. Used in cameras, drones, foldable phones. Covered in deck 09 (Flex & Rigid-Flex).

Ceramic Substrates

Highest thermal + best high-frequency option

- ▶ Al_2O_3 (alumina) — most common, 25 W/mK thermal conductivity
- ▶ AlN (aluminium nitride) — 180 W/mK, near-metal heat transfer
- ▶ BeO (beryllium oxide) — toxic, but 250+ W/mK; legacy use only
- ▶ Si_3N_4 (silicon nitride) — for IGBT modules, EV inverters
- ▶ Used in: high-power semis (LED, RF power amp), space, mmWave
- ▶ Costs 10-50× FR-4 — small substrates (1-5 cm²) typical, not large boards
- ▶ Manufacturing: thick-film printed conductors OR thin-film sputtered Cu/Au

Metal-Core PCBs (MCPCB)

Aluminium / Copper base for LED & power apps

- ▶ Aluminium base (most common, 1.0-2.0 mm thick)
- ▶ Dielectric layer (75-200 μm) — thermally conductive epoxy
- ▶ Copper trace layer (1-3 oz typical)
- ▶ Thermal conductivity: 1-3 W/mK for dielectric (vs 0.3 for FR-4)
- ▶ Used in: LED lighting bars, motor drivers, solar inverters
- ▶ Cost: 1.5-3 $\times$ FR-4

Why metal-core

Surface-mount power LED can dissipate 5-10 W into the board. FR-4 traps the heat $\rightarrow$ LED dies fast. Metal-core sinks heat into the aluminium $\rightarrow$ 5-10 $\times$ longer LED life.

Trade-offs

Single-layer only (you can't have a layer ON TOP of metal). Difficult to drill through-holes. Cost premium. Limited fabs.

Copper Foil Types

ED (electrodeposited) vs RA (rolled-annealed) Cu

ED vs RA copper foil

ED — Electrodeposited

- ▶ Made by electroplating Cu onto a drum
- ▶ Anisotropic grain structure
- ▶ Cheaper, more available
- ▶ Default for rigid PCBs
- ▶ Brittle in dynamic flex
- ▶ Sub- μm surface roughness

RA — Rolled-Annealed

- ▶ Made by rolling cast Cu billets then annealing
- ▶ Uniform grain — better ductility
- ▶ More expensive
- ▶ Used in dynamic flex circuits
- ▶ Survives 100k+ bend cycles
- ▶ Smoother surface (better for RF)

Substrate Selection Workflow

5-step process for any new design

- ▶ 1. Maximum operating frequency? → If ≤ 3 GHz, FR-4 works. If > 3 GHz, consider Rogers.
- ▶ 2. Maximum operating temperature? → If ≤ 105 °C, standard FR-4. If > 125 °C, High-Tg or polyimide.
- ▶ 3. Heat dissipation needs? → If > 5 W in a small area, consider metal-core or ceramic.
- ▶ 4. Mechanical requirements? → Rigid → FR-4. Flexible → polyimide. Rigid-flex → both.
- ▶ 5. Volume + cost? → Default to cheapest material that meets all above. Don't over-spec.

Cost & Availability Matrix

What's easy to buy in India / China

Material	Lead time (proto)	Min order	Indian sources
Standard FR-4	1-3 days	1 board	JLCPCB, PCBPower, Centum, Genus
High-Tg FR-4	3-5 days	1 board	Same as above
Halogen-Free FR-4	5-7 days	1 board	Selected fabs (specify in quote)
Polyimide (flex)	10-20 days	1 board	Multech, Flex-Hub, Streamline
Rogers RO4350B	10-15 days	20 boards	Few fabs — Tessolve, AT&S India
Ceramic (Al ₂ O ₃)	30-45 days	100 boards	Specialty — usually imported
Metal-core (Al)	5-10 days	10 boards	LED-fab specialists

Storage & Handling

Materials degrade over time — handle accordingly

Cu-clad laminate

Vacuum-sealed bag
6 months at 25 °C
Moisture-sensitive

Prepreg

Refrigerated 4 °C
Max 6 months
Avoid moisture pickup

Polyimide foil

Sealed bag
12 months
Light-sensitive (UV)

Rogers materials

Climate-controlled
12 months
Protect surface finish

Bare PCBs (finished)

ENIG/OSP — vacuum bag
6-12 months shelf life
OSP shortest

Moisture-sensitive devices

MSL ratings 1-6
Dry before reflow
Floor life limited

Common Material Selection Mistakes

What review catches early

- ▶ Using standard Tg FR-4 with lead-free reflow → delamination, board lifts
- ▶ Specifying Rogers for a sub-1 GHz design → 5-10× material cost for no benefit
- ▶ Using FR-4 for high-power LEDs → LED dies in 1000 hrs (vs 50k+ on metal-core)
- ▶ Forgetting halogen-free for EU enclosed-product compliance → certification fail
- ▶ Designing on Rogers without confirming local fab stocks the material
- ▶ Mixing FR-4 + Rogers in a single hybrid stackup without controlled-impedance review
- ▶ Specifying ED copper for dynamic flex application → cracks within months
- ▶ Buying material in bulk + storing badly → ruined prepreg, lifted Cu foils

Best Practices

Picks that work 95% of the time

- ▶ Default: Standard FR-4 + 1-oz Cu
- ▶ Lead-free assembly: ALWAYS High-Tg (170+)
- ▶ Industrial > 85 °C: High-Tg + halogen-free
- ▶ Wi-Fi / BT (≤ 6 GHz): FR-4 is fine
- ▶ 5G mmWave: Rogers RO4350B minimum
- ▶ Power LEDs: metal-core almost always
- ▶ Flex circuits: polyimide + RA copper
- ▶ Specify Tg, Cu weight, finish on every order
- ▶ Talk to your fab — what do they stock?
- ▶ Hybrid stackups: confirm fab capability
- ▶ Validate with material data sheets
- ▶ Order extra material for cycle test / spares

Key Takeaways

Key takeaways from this lecture

FR-4 = workhorse

95% of designs.

Tg = lead-free survival

High-Tg minimum for modern assembly.

Rogers for > 3 GHz

mmWave needs PTFE-class.

Polyimide = flex + heat

260 °C continuous.

Metal-core for LEDs

Heat sink built in.

Don't over-spec

Cost scales 10-50× past FR-4.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ IPC-4101 — Specification for laminate materials
- ▶ Rogers Corporation datasheets — rogerscorp.com
- ▶ DuPont Pyralux design guide — polyimide flex
- ▶ Henkel Bergquist app notes — metal-core PCBs
- ▶ Isola, Nelco, Panasonic Megtron — alternative high-speed laminates
- ▶ JLCPCB / PCBWay material lists — what's actually available
- ▶ Indian Printed Circuit Association (IPCA) — local supplier directory
- ▶ Phil's Lab — material selection videos

Thank you • Build something this week.